

1D CNN — Architecture Variant Comparison

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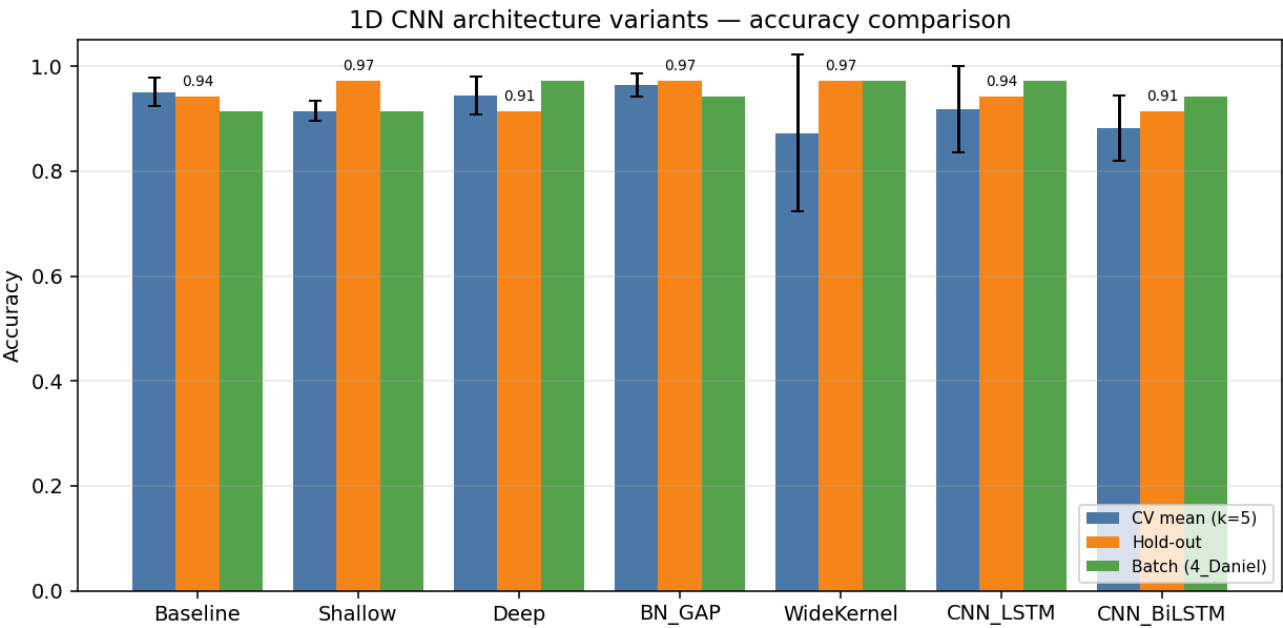
Data & configuration

Training path 1	/home/jestin/ThesisRepo/ML/NewTestData/1_Alan/Dynamic
Training path 2	/home/jestin/ThesisRepo/ML/NewTestData/2_Alex/Dynamic
Training path 3	/home/jestin/ThesisRepo/ML/NewTestData/3_Anghad/Dynamic
Training path 4	/home/jestin/ThesisRepo/ML/NewTestData/6_Harry/Dynamic
Training path 5	/home/jestin/ThesisRepo/ML/NewTestData/8_Jestin/Dynamic
Training path 6	/home/jestin/ThesisRepo/ML/NewTestData/11_Mansh/Dynamic
Training path 7	/home/jestin/ThesisRepo/ML/NewTestData/12_Marcus/Dynamic
Training path 8	/home/jestin/ThesisRepo/ML/NewTestData/14_StephenV2/Dynamic
Training path 9	/home/jestin/ThesisRepo/ML/NewTestData/15_Tash/Dynamic
Training path 10	/home/jestin/ThesisRepo/ML/NewTestData/17_Raquel/Dynamic
Batch-test root	/home/jestin/ThesisRepo/ML/NewTestData/4_Daniel/Dynamic
Total trials	341 (train pool 306, hold-out 35)
Classes	7: Double_Lower, Double_Nothing, Double_Pistol_Recoil, Double_Raise, Double_Wiggle, Double_cmere, Drum_Roll
Sensor channels	176 (sequence length 90)
Resample / filter	90 steps · Butterworth cutoff 15.0 Hz, order 4
Normalisation	minmax
CV folds · shuffle	5 · True
Augmentation	on · copies=5 · amplitude_scale, baseline_offset
Epochs · batch size	15 · 16

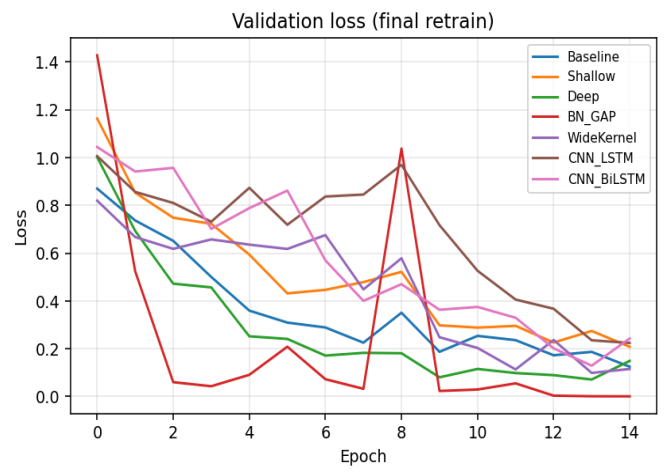
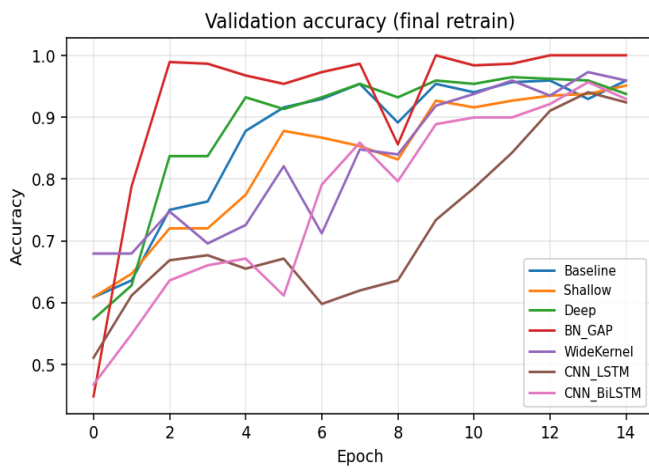
Variant comparison (summary)

Variant	Params	CV mean (k=5)	CV std	Hold-out acc	Hold-out loss	Batch acc	Batch n
Baseline	113,255	0.9511	0.0271	0.9429	0.1661	0.9143	32/35
Shallow	72,487	0.9151	0.0189	0.9714	0.1627	0.9143	32/35
Deep	204,007	0.9443	0.0369	0.9143	0.1725	0.9714	34/35
BN_GAP	57,319	0.9641	0.0217	0.9714	0.1056	0.9429	33/35
WideKernel	131,687	0.8723	0.1494	0.9714	0.1231	0.9714	34/35
CNN_LSTM	68,455	0.9182	0.0830	0.9429	0.2306	0.9714	34/35
CNN_BiLSTM	105,575	0.8827	0.0621	0.9143	0.2818	0.9429	33/35

Best on hold-out: **Shallow** · Best on CV: **BN_GAP**



Training curves (final retrain on full training pool)



Variant: Baseline

Conv1D(32,k=4)→Pool→Conv1D(64,k=4)→Pool→Flatten→Dense(64)→Dropout(0.3)→Softmax

Trainable params	113,255
CV mean ± std	0.9511 ± 0.0271
Hold-out test acc / loss	0.9429 · 0.1661
Batch-test acc	0.9143 (32/35)

Cross-validation folds

Fold	Train (post-aug)	Val	Val acc	Best val	Final val
1	1464	62	0.9194	0.9516	0.9194
2	1470	61	0.9344	0.9344	0.9344
3	1470	61	0.9836	1.0000	0.9836
4	1470	61	0.9836	0.9836	0.9836
5	1470	61	0.9344	0.9672	0.9344

Per-class hold-out metrics

Class	Precision	Recall	F1	Support
Double_Lower	1.000	1.000	1.000	5
Double_Nothing	1.000	0.800	0.889	5
Double_Pistol_Recoil	1.000	1.000	1.000	5
Double_Raise	1.000	0.800	0.889	5
Double_Wiggle	0.833	1.000	0.909	5
Double_cmere	0.833	1.000	0.909	5
Drum_Roll	1.000	1.000	1.000	5
macro avg	0.952	0.943	0.942	35
weighted avg	0.952	0.943	0.942	35

Batch test per class (4_Daniel)

Class	Correct	Total	Accuracy
Double_Lower	5	5	1.000
Double_Nothing	2	5	0.400
Double_Pistol_Recoil	5	5	1.000
Double_Raise	5	5	1.000
Double_Wiggle	5	5	1.000
Double_cmere	5	5	1.000
Drum_Roll	5	5	1.000
Overall	32	35	0.914

Variant: Shallow

Single Conv block: Conv1D(32,k=5)→Pool→Flatten→Dense(32)→Dropout(0.3)→Softmax

Trainable params	72,487
CV mean ± std	0.9151 ± 0.0189
Hold-out test acc / loss	0.9714 · 0.1627
Batch-test acc	0.9143 (32/35)

Cross-validation folds

Fold	Train (post-aug)	Val	Val acc	Best val	Final val
1	1464	62	0.9032	0.9194	0.9032
2	1470	61	0.9180	0.9344	0.9180
3	1470	61	0.9344	0.9836	0.9344
4	1470	61	0.8852	0.8852	0.8852
5	1470	61	0.9344	0.9344	0.9344

Per-class hold-out metrics

Class	Precision	Recall	F1	Support
Double_Lower	1.000	1.000	1.000	5
Double_Nothing	1.000	0.800	0.889	5
Double_Pistol_Recoil	1.000	1.000	1.000	5
Double_Raise	1.000	1.000	1.000	5
Double_Wiggle	0.833	1.000	0.909	5
Double_cmere	1.000	1.000	1.000	5
Drum_Roll	1.000	1.000	1.000	5
macro avg	0.976	0.971	0.971	35
weighted avg	0.976	0.971	0.971	35

Batch test per class (4_Daniel)

Class	Correct	Total	Accuracy
Double_Lower	5	5	1.000
Double_Nothing	3	5	0.600
Double_Pistol_Recoil	5	5	1.000
Double_Raise	5	5	1.000
Double_Wiggle	4	5	0.800
Double_cmere	5	5	1.000
Drum_Roll	5	5	1.000
Overall	32	35	0.914

Variant: Deep

Three Conv blocks: Conv1D(32,4)→Conv1D(64,4)→Conv1D(128,3)→Flatten→Dense(128)→Dropout(0.4)→Softmax

Trainable params	204,007
CV mean ± std	0.9443 ± 0.0369
Hold-out test acc / loss	0.9143 · 0.1725
Batch-test acc	0.9714 (34/35)

Cross-validation folds

Fold	Train (post-aug)	Val	Val acc	Best val	Final val
1	1464	62	0.9839	0.9839	0.9839
2	1470	61	0.9016	0.9344	0.9016
3	1470	61	0.9016	0.9836	0.9016
4	1470	61	0.9508	0.9836	0.9508
5	1470	61	0.9836	0.9836	0.9836

Per-class hold-out metrics

Class	Precision	Recall	F1	Support
Double_Lower	1.000	1.000	1.000	5
Double_Nothing	0.667	0.800	0.727	5
Double_Pistol_Recoil	1.000	0.600	0.750	5
Double_Raise	1.000	1.000	1.000	5
Double_Wiggle	0.833	1.000	0.909	5
Double_cmere	1.000	1.000	1.000	5
Drum_Roll	1.000	1.000	1.000	5
macro avg	0.929	0.914	0.912	35
weighted avg	0.929	0.914	0.912	35

Batch test per class (4_Daniel)

Class	Correct	Total	Accuracy
Double_Lower	5	5	1.000
Double_Nothing	4	5	0.800
Double_Pistol_Recoil	5	5	1.000
Double_Raise	5	5	1.000
Double_Wiggle	5	5	1.000
Double_cmere	5	5	1.000
Drum_Roll	5	5	1.000
Overall	34	35	0.971

Variant: BN_GAP

Conv+BN+ReLU ×3 with GlobalAveragePooling1D head — fewer params, less overfit

Trainable params	57,319
CV mean ± std	0.9641 ± 0.0217
Hold-out test acc / loss	0.9714 · 0.1056
Batch-test acc	0.9429 (33/35)

Cross-validation folds

Fold	Train (post-aug)	Val	Val acc	Best val	Final val
1	1464	62	0.9516	0.9677	0.9516
2	1470	61	0.9344	0.9344	0.9344
3	1470	61	1.0000	1.0000	1.0000
4	1470	61	0.9672	0.9672	0.9672
5	1470	61	0.9672	0.9836	0.9672

Per-class hold-out metrics

Class	Precision	Recall	F1	Support
Double_Lower	1.000	1.000	1.000	5
Double_Nothing	1.000	0.800	0.889	5
Double_Pistol_Recoil	1.000	1.000	1.000	5
Double_Raise	1.000	1.000	1.000	5
Double_Wiggle	0.833	1.000	0.909	5
Double_cmere	1.000	1.000	1.000	5
Drum_Roll	1.000	1.000	1.000	5
macro avg	0.976	0.971	0.971	35
weighted avg	0.976	0.971	0.971	35

Batch test per class (4_Daniel)

Class	Correct	Total	Accuracy
Double_Lower	5	5	1.000
Double_Nothing	3	5	0.600
Double_Pistol_Recoil	5	5	1.000
Double_Raise	5	5	1.000
Double_Wiggle	5	5	1.000
Double_cmere	5	5	1.000
Drum_Roll	5	5	1.000
Overall	33	35	0.943

Variant: WideKernel

Conv1D(32,k=8)→Pool→Conv1D(64,k=8)→Pool→Flatten→Dense(64)→Dropout(0.3)→Softmax

Trainable params	131,687
CV mean ± std	0.8723 ± 0.1494
Hold-out test acc / loss	0.9714 · 0.1231
Batch-test acc	0.9714 (34/35)

Cross-validation folds

Fold	Train (post-aug)	Val	Val acc	Best val	Final val
1	1464	62	0.9516	0.9516	0.9516
2	1470	61	0.5738	0.9180	0.5738
3	1470	61	0.9508	0.9672	0.9508
4	1470	61	0.9508	0.9672	0.9508
5	1470	61	0.9344	0.9672	0.9344

Per-class hold-out metrics

Class	Precision	Recall	F1	Support
Double_Lower	1.000	1.000	1.000	5
Double_Nothing	1.000	0.800	0.889	5
Double_Pistol_Recoil	1.000	1.000	1.000	5
Double_Raise	1.000	1.000	1.000	5
Double_Wiggle	0.833	1.000	0.909	5
Double_cmere	1.000	1.000	1.000	5
Drum_Roll	1.000	1.000	1.000	5
macro avg	0.976	0.971	0.971	35
weighted avg	0.976	0.971	0.971	35

Batch test per class (4_Daniel)

Class	Correct	Total	Accuracy
Double_Lower	5	5	1.000
Double_Nothing	4	5	0.800
Double_Pistol_Recoil	5	5	1.000
Double_Raise	5	5	1.000
Double_Wiggle	5	5	1.000
Double_cmere	5	5	1.000
Drum_Roll	5	5	1.000
Overall	34	35	0.971

Variant: CNN_LSTM

Conv1D(32,4)→Pool→Conv1D(64,4)→Pool→Dropout→LSTM(64)→Dense(64)→Dropout→Softmax

Trainable params	68,455
CV mean ± std	0.9182 ± 0.0830
Hold-out test acc / loss	0.9429 · 0.2306
Batch-test acc	0.9714 (34/35)

Cross-validation folds

Fold	Train (post-aug)	Val	Val acc	Best val	Final val
1	1464	62	0.9355	0.9516	0.9355
2	1470	61	0.7541	0.9016	0.7541
3	1470	61	0.9672	0.9672	0.9672
4	1470	61	0.9672	0.9672	0.9672
5	1470	61	0.9672	0.9672	0.9672

Per-class hold-out metrics

Class	Precision	Recall	F1	Support
Double_Lower	1.000	1.000	1.000	5
Double_Nothing	1.000	0.800	0.889	5
Double_Pistol_Recoil	1.000	1.000	1.000	5
Double_Raise	1.000	1.000	1.000	5
Double_Wiggle	1.000	0.800	0.889	5
Double_cmere	1.000	1.000	1.000	5
Drum_Roll	0.714	1.000	0.833	5
macro avg	0.959	0.943	0.944	35
weighted avg	0.959	0.943	0.944	35

Batch test per class (4_Daniel)

Class	Correct	Total	Accuracy
Double_Lower	5	5	1.000
Double_Nothing	4	5	0.800
Double_Pistol_Recoil	5	5	1.000
Double_Raise	5	5	1.000
Double_Wiggle	5	5	1.000
Double_cmere	5	5	1.000
Drum_Roll	5	5	1.000
Overall	34	35	0.971

Variant: CNN_BiLSTM

Conv1D(32,4)→Pool→Conv1D(64,4)→Pool→Dropout→BiLSTM(64)→Dense(64)→Dropout→Softmax

Trainable params	105,575
CV mean ± std	0.8827 ± 0.0621
Hold-out test acc / loss	0.9143 · 0.2818
Batch-test acc	0.9429 (33/35)

Cross-validation folds

Fold	Train (post-aug)	Val	Val acc	Best val	Final val
1	1464	62	0.7742	0.7742	0.7742
2	1470	61	0.9344	0.9344	0.9344
3	1470	61	0.8525	0.9508	0.8525
4	1470	61	0.9180	0.9180	0.9180
5	1470	61	0.9344	0.9508	0.9344

Per-class hold-out metrics

Class	Precision	Recall	F1	Support
Double_Lower	1.000	1.000	1.000	5
Double_Nothing	1.000	0.600	0.750	5
Double_Pistol_Recoil	1.000	1.000	1.000	5
Double_Raise	1.000	1.000	1.000	5
Double_Wiggle	0.625	1.000	0.769	5
Double_cmere	1.000	1.000	1.000	5
Drum_Roll	1.000	0.800	0.889	5
macro avg	0.946	0.914	0.915	35
weighted avg	0.946	0.914	0.915	35

Batch test per class (4_Daniel)

Class	Correct	Total	Accuracy
Double_Lower	5	5	1.000
Double_Nothing	4	5	0.800
Double_Pistol_Recoil	5	5	1.000
Double_Raise	5	5	1.000
Double_Wiggle	5	5	1.000
Double_cmere	4	5	0.800
Drum_Roll	5	5	1.000
Overall	33	35	0.943